

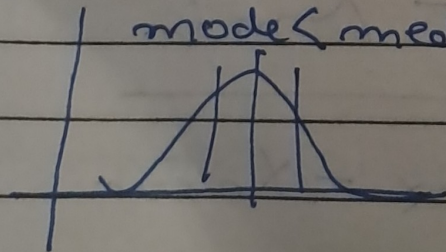
Positively skewed distribution

What is In positively skewed distribution, most values are concentrated on the lower end with a long tail extending to the right. A few high values pull the average to the right of the median.

Skewness ÷ A distribution or asymmetry that deviates from the symmetrical bell curve

also called right-skewed or right-tailed

mode < median < mean



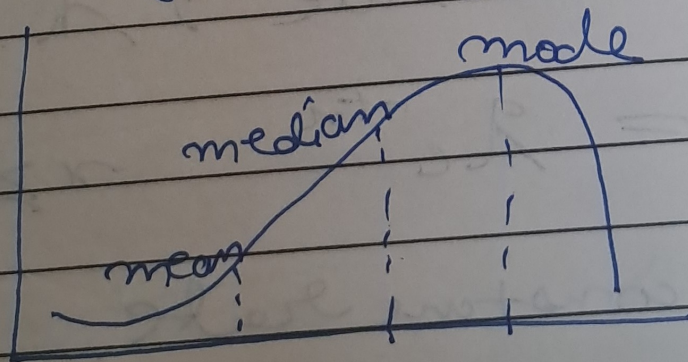
when ÷ useful for data with rare but significant high values such as income levels where few

individual earn much more than the rest.

The mean is greater than the median due to outlier on the higher end.

Negatively skewed distribution.

most values are clustered at the higher end, with a few values creating a long tail to the left.



mode > median > mean

used for dataset where values are typically higher but a few lower values exist (returning)

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The mean is generally less than median due to leftward skewness.

It helps detect outliers (cases which are generally high but occasionally much lower).

Exponential Distribution:

It describes the time between event in a process where the event occur independently at a constant rate λ .

$$f(x) = \lambda e^{-\lambda x} \quad x \geq 0$$

λ = constant rate

Bernoulli Distribution

It means mode is a single ~~experi~~ experiment with two possible outcomes.

Success ($X=1$) ; failure ($X=0$)

Binomial distribution (Discrete)

Binomial dist extends Bernoulli to n independent trials.

$$P(X=K) \Rightarrow \binom{n}{K} p^K (1-p)^{n-K}$$

X = random variable (no of success out of n trials)

n = total no of trial (100)

K = No of success ($0 \leq K \leq n$)

P = p (Success in 1 trial) $0 \leq p \leq 1$

$$\binom{n}{K} = \frac{n!}{(n-K)! K!}$$

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Uniform distribution: (continuous)

$$f(x) = \frac{1}{b-a}$$

$$a \leq x \leq b$$

The probability of any value within the range $[a, b]$ is same.

Uniform Distribution (discrete)
all outcomes are equally likely.

$$P(x) = \frac{1}{n}$$

→ total no.

Confidence Interval

$\bar{x} = 50 \rightarrow$ point estimate

It is a range of values within which we expect a particular parameter to fall.

Confidence Interval = point estimate $\pm$ margin of error

Hypothesis

Hypothesis testing :-

A statistical hypothesis is a statistical interference used to decide whether the data at hand is sufficient to support a particular hypothesis.

Hypothesis testing allows us to make probability statement about population parameters.

Null hypothesis (H_0)

The null hypothesis assumes that there is no significant relationship or effect between two variables. [In simple terms, it says nothing new is happening.]

It serves as a starting point. H_0 represents the static quo, the assumption of no effect until proven otherwise.

The purpose of HT is to gather evidence to reject or fail null hypothesis in favour of alternate hypothesis, which claims there is significant effect or relationship.

Alternate Hypothesis -

H_a or H_1 It is a statement that contradicts the H_0 & claims there is significance effect or relationship.

Rejection Region Method :-

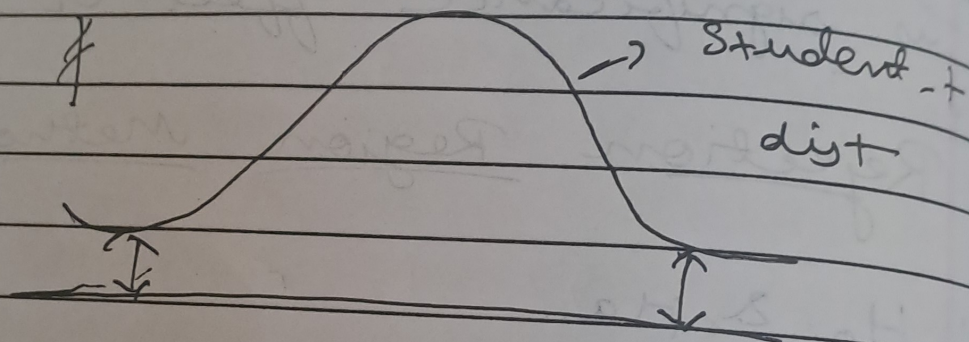
- 1) H_0 & H_a
- 2) α - Value
- 3) assumption
- 4) decide test
 ↳ Z test, t-test
- 5) Value
- 6) Test conduct
- 7) Reject / Accept
- 8) Stat Result

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$P < \alpha \rightarrow H_0 \text{ rejected}$
 $P > \alpha \rightarrow H_0 \text{ accept}$

P-value $\rightarrow$ It is a measure of the strength of the evidence against the null hypothesis.

T-test



One sample t-test compares the mean of a single sample to a known μ .

$$t = \frac{\bar{x} - \mu}{s\sqrt{n}}$$

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$$df = \text{degree of freedom}$$
$$= n - 1$$

2. Independent Two sample t-test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

3.) Paired t-test

$$t = \frac{\bar{d}}{sd / \sqrt{n}}$$

$\bar{d} = 2 \rightarrow \text{mean diff}$

$sd = \text{std-diff}$

Chi-Square test

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O $\rightarrow$ observed frequency

E $\rightarrow$ Expected frequency

$$E = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

$$df = (r-1) \times (c-1)$$

r = no of row

c = no of columns.

Kurtosis

Kurtosis measures "tailedness" of a distribution or how extreme the outliers are.

1. Mesokurtic

- Tails are similar to NP
- distribution with kurtosis ≈ 3
- eg Standard Normal distribution

2. Leptokurtic

- a distribution with kurtosis > 3
- Heavy tails (with more/extreme outliers)

eg: t -distribution with very small df .

3.) Platykurtic

- distribution with kurtosis
- light tails (fewer extreme outliers)
- eg - uniform dist

Excess kurtosis $\rightarrow K - 3$

$K > 0 \rightarrow \text{Leptokurtic}$

$K < 0 \rightarrow \text{Platykurtic}$

$$K = \frac{n \cdot \sum (x_i - \bar{x})^4}{(\sum (x_i - \bar{x})^2)^2} \cdot \frac{1}{n}$$

n = no of observation
 x_i = Each data point

$\bar{x}$ = mean

High kurtosis (Leptokurtic)

- more extreme outliers
- higher likelihood of rare, extreme values.

eg: ~~the~~ financial returns during market crisis/crashes

Lower kurtosis (Platykurtic)

- fewer extreme outliers
- data is evenly spread

Kurtosis near 3 (mesokurtic)
similar to normal distribution
leptokurtic distribution

normal distribution

Platykurtic

distribution

